

MEP Drawing Reading Guide

LUA BIM LABS Starter - Track 3 Quick Reference | Track completion card - Day 21

Drawing Types Reference

Drawing Type	Definition	Created By	Primary Use
Design Drawing	Shows the engineer's design intent - system layout, sizes, and specification	Design Engineer	Basis for BIM modeling and procurement
Coordination Drawing	Shows clash-resolved MEP routing	BIM Coordinator	Site installation and setting-out
Shop Drawing	Shows fabrication level detail for construction	Contractor / Fabricator	Off-site manufacturing (spools, panels)
As-Built Drawing	Records the actual installed condition - deviations from design captured	Contractor (BIM-assisted)	Facility management and handover

MEP Plan Reading Tips

HVAC Plans - what to check:

- Supply air routes from AHU to diffusers (confirm system assignment on each duct)
- Return air routes from grilles back to AHU (do not confuse with supply)
- VAV box locations and connected branch ducts
- Equipment tags (AHU, FCU, EF) matching the equipment schedule
- Duct size annotations at key routing changes

Piping Plans - what to check:

- Flow direction arrows on supply and return lines
- Pipe size annotations - confirm they match the design specification
- Valve symbols - isolation valves, check valves, pressure regulators
- Equipment connection points (pumps, coils, heat exchangers)
- Drain and vent pipe routes for sanitary systems (look for slope notation)

Electrical Plans - what to check:

- Cable tray routing from main switchboard to distribution boards
- Panel board locations matching the single-line diagram
- Circuit homerun lines from panels to major equipment
- Conduit routes and junction box positions
- Load labels on distribution equipment

Schematic Symbols Reference

System	Common Symbols
HVAC Supply Air	Solid rectangle duct in plan; arrow showing flow direction
HVAC Return Air	Dashed rectangle duct in plan; arrow reversed toward AHU
Chilled Water	CWS (supply) and CWR (return) with flow arrows
Hot Water	HWS and HWR labeled, with isolating valve symbols
Drainage (gravity)	Dashed line with slope notation (e.g., 1:50)
Sprinkler	Solid pipe line with "X" or dot symbols for heads
Cable Tray	Double-line rectangle in plan; size labeled (e.g., 600x100)
Conduit	Single line with size callout; homerun arrows to panel

Coordination vs. Design Drawing Differences

	Design Drawing	Coordination Drawing
Purpose	Show design intent	Show clash-resolved routing for installation
LOD	LOD 200-300	LOD 350
Elevation data	Approximate or not shown	Confirmed pipe inverts and duct soffit levels
Structural reference	May not include beams	Structural grid and key beams shown
Clash status	Not applicable	All Major clashes resolved before issue
Issued by	Design Engineer	BIM Coordinator
Used by	Modeling team, QA	Site installation teams